

Ekow Nii Saki Sackey

Software Engineer · Backend · Cloud · Quality Assurance

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PROFILE

Software engineer with a background in Computer Engineering. I have professional Software Engineering experience spanning full-stack web development internships and most recently an intensive one-year professional training programme in Java, Spring Boot, AWS Architecture and Software Quality. I am dedicated and self-driven, a quick learner with strong communication skills and an ability to adapt quickly to change. I am technically versatile and willing to deepen my expertise while contributing meaningful work to any Software project.

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, SQL, Go

Backend: Spring Boot, Spring Data JPA, Spring Security, Spring Cloud (Eureka, Gateway), RabbitMQ, Resilience4j, REST, GraphQL, Flask, FastAPI, Node.js, Express.js

Frontend: Vue 3, Pinia, Vue Router, Vite, React, Next.js, Tailwind CSS

Databases: PostgreSQL, MongoDB, DynamoDB, Redis

Cloud and DevOps: AWS (Cloud Practitioner Certified) - ECS/Fargate, Lambda, API Gateway, S3, CloudFront, RDS and RDS Proxy, DynamoDB, EventBridge, SNS/SQS, Cognito, Amplify, VPC, IAM, CloudFormation, SAM; Docker, GitHub Actions, CI/CD, OIDC federation, Infrastructure as Code

Testing and Quality: JUnit 5, Vitest, JMeter, VisualVM, test planning, traceability, defect management

AI and ML: RAG pipelines, Hugging Face Transformers, model fine-tuning

Tools: Git, Figma, Postman

EXPERIENCE

Software Engineering Trainee (Backend · Cloud · Quality Assurance)

AmaliTech - Accra, Ghana · *November 2025 - September 2026*

- Delivered an internal data validation platform as part of a five-engineer Scrum team: volunteered into frontend, then into quality engineering.
- Served as the project's sole quality gate: raised and drove 58 defects to resolution across five epics, reaching handover with no major defects or blockers outstanding.
- Rebuilt the backend test infrastructure, taking the suite from 296 tests to 445, exercising every main path against Testcontainers PostgreSQL, a real SMTP server and a real browser.
- Authored the product specification after an undocumented mid-project pivot and drove it to Product Owner sign-off.
- Profiled and optimised API performance under load using async patterns, concurrency control and caching, with before-and-after measurements taken in VisualVM and JMeter.
- Delivered multiple AWS-centred lab projects as working deployments within the Cloud Computing training curriculum: ECS Fargate with blue/green CI/CD, multi-AZ VPCs in CloudFormation, RDS with connection proxying and Redis caching, and event-driven serverless on Lambda, DynamoDB and EventBridge.

Full-Stack Developer Intern

Mon & Associates - Remote · *February 2024 - August 2024*

- Transformed AllAcademies, a learning and automated assessment platform, from a bare-bones skeleton interface into a polished, intuitive product, owning the full UI/UX redesign end to end.
- Built responsive frontend interfaces using Tailwind CSS and GSAP animations, prototyped in Figma ahead of implementation, and contributed to backend development in PHP and Laravel.

Python Development Intern

GlobalCode - Accra, Ghana · *September 2023 - October 2023*

- Went properly into the weeds with Python for the first time across an intensive one-month bootcamp, moving from algorithm scripting and data handling into full-stack web development with Flask and Bootstrap.
- Learned to integrate and consume third-party REST APIs, and to reason about how data moves between a service, a backend and a rendered page. Gained hands-on experience of the vast capabilities of the Python programming language through Raspberry Pi projects.
- Capstone project: built MEJ News with two other developers as the programme's final team deliverable - a news aggregator integrating a third-party news API to fetch, sort and categorise articles dynamically, in Flask with a Bootstrap frontend.

PROJECTS

Lab Results Validation App - Full-Stack · AmaliTech Internal Platform · Jun 2026 - Sep 2026

- Internal platform that ingests instructors' grading workbooks from SharePoint, validates them against reference data and maintains a full audit trail. Vue 3 and Pinia frontend against a Spring Boot and PostgreSQL backend.
- Shipped the entire admin and instructor surface as sole frontend engineer: 16 Vue 3 and TypeScript screens with role-based access control, plus a design system of tokens and primitives.
- Moved into the sole QA role on the same product and diagnosed its only blocker: change detection keyed record identity on a mutable field, so ordinary re-grades were silently discarded and same-day duplicates dropped entirely. Traced both failure paths and proposed the fix that shipped with a database migration.
- Made the product testable without its external dependency by refactoring the SharePoint integration behind an interface with a fixture-backed implementation, and added a 17-journey Playwright regression pack.

Food Delivery Platform - Microservices · Java, Spring Cloud, RabbitMQ, Docker

- Decomposed a monolithic Spring Boot backend into four independently deployable microservices, each with its own database, replacing cross-domain joins with service-to-service REST calls.
- Connected them with Eureka service discovery, a Spring Cloud Gateway handling routing, JWT authentication and rate limiting, RabbitMQ event messaging and Resilience4j circuit breakers with fallbacks; the full system up and running from a single Docker Compose command.

Serverless Task Manager - AWS SAM · Lambda, DynamoDB, EventBridge, Cognito

- Fully serverless, event-driven application: Cognito authentication, CRUD through API Gateway and Lambda onto DynamoDB, and a frontend hosted on Amplify.
- Implemented per-task scheduled expiry through EventBridge with SNS email notification, and idempotent cancellation through DynamoDB Streams into an SQS FIFO queue; defined entirely as infrastructure as code with an automated deployment pipeline.

TwigPT chatbot - Final Year Project · Dec 2024 - Sep 2025

- Developed a generative language model based on the GPT architecture, fine-tuned entirely on Twi, a low-resource African language; conducted the NLP research under the supervision of a Senior Lecturer and built the data collection and training pipelines from the JW corpus and transcribed local news sources.
- Built the serving layer and chat application around it: a single streaming interface hiding three different execution paths, namely a local Ollama server, Hugging Face Transformers loaded in-process, and a remote Gradio Space.
- Designed a feedback loop that turned ordinary use of the application into labelled training data for the next iteration of the model.
- Research thesis graded A.

EDUCATION

BSc Computer Engineering - 2021 - 2025

University of Ghana

Second Class Honours (Upper Division)

CERTIFICATIONS

- **AWS Certified Cloud Practitioner** - Amazon Web Services
- **Generative AI Fundamentals** - Databricks
- **Responsive Web Design** - freeCodeCamp
- **Scientific Computing with Python** - freeCodeCamp

LANGUAGES

English · Ga · Twi